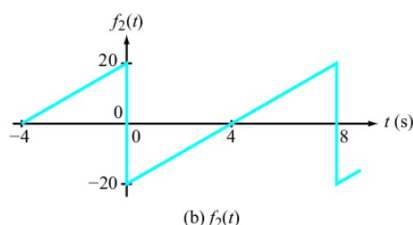
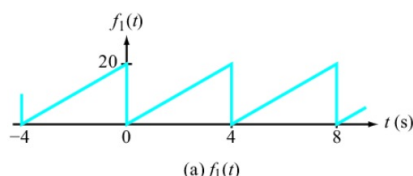


Zybooks extra exercises from 5.4 sine/cosine of Fourier Series

EXERCISE 5.4.1



(a) The Fourier series of the periodic waveform shown in figure (a) above is

$$f_1(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi t}{2}\right).$$

Determine the Fourier series of waveform $f_2(t)$ in figure (b).

$f_2(t)$ is a period $T_0=4s$ ± 20 at multiples of T

6. Sawtooth $x(t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2A}{n\pi} \sin\left(\frac{2n\pi t}{T_0}\right)$

$A=20, T_0=4$

↑ formula we start with

$t=0 \rightarrow t=\frac{T_0}{2} = \frac{4}{2} = 2$; $f_2(t) = s(t-2)$

$f_2(t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{40}{n\pi} \sin\left(\frac{n\pi}{2}(t-2)\right)$ * $\sin(n\pi) = 0$

$\sin\left(\frac{n\pi}{2}(t-2)\right) = \sin\left(\frac{n\pi}{2}t - \frac{n\pi}{2} \cdot 2\right) = \sin\left(\frac{n\pi}{2}t - n\pi\right)$

$\sin\left(\frac{n\pi}{2}t - n\pi\right) = \sin\left(\frac{n\pi}{2}t\right)\cos(n\pi) - \cos\left(\frac{n\pi}{2}t\right)\sin(n\pi) = (-1)^n \sin\left(\frac{n\pi}{2}t\right)$

* above we use angle subtraction identity: $\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$
here $A = \frac{n\pi}{2}t$ and $B = n\pi$, $\cos(n\pi) = (-1)^n$ and $\sin(n\pi) = 0$

combine: $f_2(t) = \sum_{n=1}^{\infty} (-1)^{n+1} (-1)^n \frac{40}{n\pi} \sin\left(\frac{n\pi}{2}t\right) = \sum_{n=1}^{\infty} (-1)^{2n+1} \frac{40}{n\pi} \sin\left(\frac{n\pi}{2}t\right)$

* $(-1)^{2n+1} = -1$ for every n ; alternating sign becomes constant

$f_2(t) = -\frac{40}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}t\right)$

EXERCISE 5.4.2

Let $x(t)$ be a real-valued and periodic signal with period T_0 sec and maximum frequency B Hz. The signal has a nonzero component at exactly B Hz. $f_0 = \frac{1}{T_0}$ [Hz]

(a) Show that $x(t)$ can be expressed using an expression with only $(2BT_0 + 1)$ constants.

(b) $x(t)$ has period 0.1 s and maximum frequency 20 Hz. Known values are:

$\left. \begin{array}{l} x(0.00) = 5, \\ x(0.02) = 1, \\ x(0.04) = 3, \\ x(0.06) = 2, \\ x(0.08) = 4. \end{array} \right\} 0.02 \text{ sec} = \frac{T_0}{5}$

Determine an explicit formula for $x(t)$.

Part A:

basic trig Fourier: $x(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n t}{T_0}\right) + b_n \sin\left(\frac{2\pi n t}{T_0}\right) \right]$

max frequency is B [Hz]; $N f_0 = B \rightarrow N = BT_0 = n$ nonzero

$x(t) = a_0 + \sum_{n=1}^N a_n \cos\left(\frac{2\pi n t}{T_0}\right) + \sum_{n=1}^N b_n \sin\left(\frac{2\pi n t}{T_0}\right)$

$a_0: 1; a_n: N; b_n: N$ constraints Total = $1 + N + N = 1 + 2N = 2BT_0 + 1$

Part B:

$T_0 = 0.1, B = 20 \text{ [Hz]} \rightarrow N = BT_0 = 20 \cdot 0.1 = 2$; # unknown constants = $2(2) + 1 = 5$

$x(t) = a_0 + a_1 \cos(\omega_0 t) + a_2 \cos(2\omega_0 t) + b_1 \sin(\omega_0 t) + b_2 \sin(2\omega_0 t)$; $\omega_0 = \frac{2\pi}{T_0} = \frac{20\pi}{0.1}$

plug in $t_m = 0, 0.02, 0.04, 0.06, 0.08 \rightarrow \omega_0 t_m = \frac{2\pi m}{5}$

$a_0 = \frac{1}{5} \sum_{m=0}^4 x(t_m) = \frac{5+1+3+2+4}{5} = 3$

$a_n = \frac{2}{5} \sum_{m=0}^4 x(t_m) \cos\left(\frac{2\pi n m}{5}\right)$ $b_n = \frac{2}{5} \sum_{m=0}^4 x(t_m) \sin\left(\frac{2\pi n m}{5}\right)$

$a_1 = 1, a_2 = 1$ $b_1 = -\sqrt{\frac{25-2\sqrt{5}}{25}} = -0.9062, b_2 = -\sqrt{\frac{25+2\sqrt{5}}{25}} = -1.0858$

answer: $x(t) = 3 + \cos(20\pi t) + \cos(40\pi t) - 0.9062 \sin(20\pi t) - 1.0858 \sin(40\pi t)$

double verify by plugging in!

